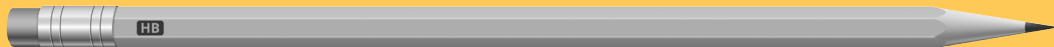


Colorado Measures of Academic Success



Grade 7 Mathematics



Paper Practice Resource for Students

ITEM SET 1 - SECTION 1 (Non-Calculator)

Directions:

This Item Set has two sections: a non-calculator section and a calculator section. You will now take the non-calculator section. You may not use a calculator.

1. Which expression is equivalent to $\frac{1}{2} - \frac{3}{5}$?

Ⓐ $-\frac{3}{5} + \left(-\frac{1}{2}\right)$

Ⓑ $-\frac{1}{2} + \left(\frac{3}{5}\right)$

Ⓒ $\frac{1}{2} + \left(-\frac{3}{5}\right)$

Ⓓ $\frac{3}{5} + \left(\frac{1}{2}\right)$

2. In 1 year, the water level of a lake changes by $-\frac{3}{8}$ inch.

If the water level of the lake continues to change at this rate for 7 years, how many inches will the water level of the lake have changed?

Enter your answer in the space provided. Enter **only** your answer.

3. In one hour, Earth travels approximately 67,000 miles in its orbit around the Sun.

Which equation represents the relationship between the total number of miles, m , that Earth travels in h hours?

- (A) $67,000 + m = h$
- (B) $67,000 + h = m$
- (C) $67,000m = h$
- (D) $67,000h = m$

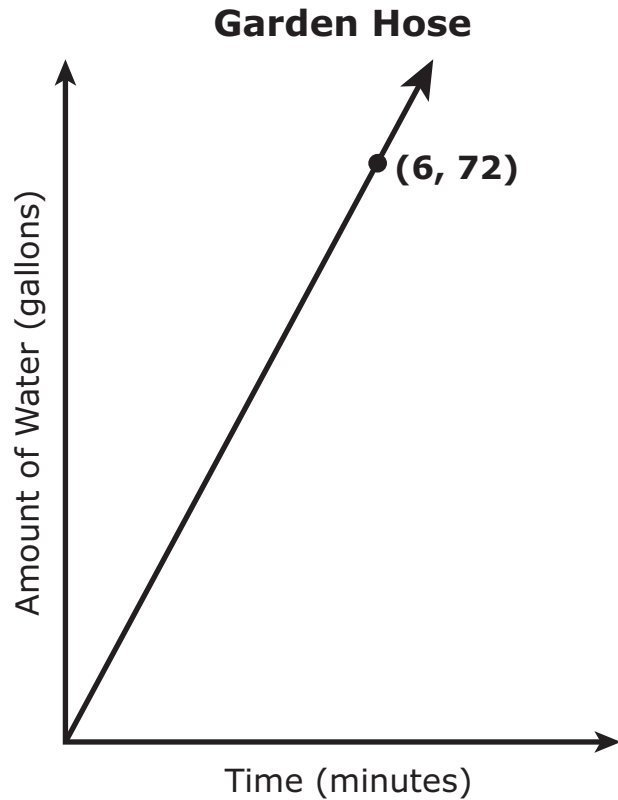
4. Create an expression that has the same value as $(6x - 4) + (x + 5)$.

Write the correct numbers from the list in the blank boxes. Each number may be used once, more than once, or not at all.

1	2	4	5	6	7	9	11
---	---	---	---	---	---	---	----

	$\times +$	
--	------------	--

5. The graph represents the amount of water, in gallons, y , that flows through a garden hose in x minutes.



Which statement correctly describes the meaning of the point shown on this graph?

- Ⓐ Every 6 minutes, 72 gallons of water flow through the hose.
- Ⓑ Every 72 minutes, 6 gallons of water flow through the hose.
- Ⓒ Every 6 minutes, 12 gallons of water flow through the hose.
- Ⓓ Every 12 minutes, 6 gallons of water flow through the hose.

6. Store A has meat on sale for $1\frac{1}{2}$ lbs for \$5.34. The table shows the prices for meat at three nearby stores.

Place a check mark (✓) to identify how the price of the meat at each nearby store compares to the price of the meat at Store A. Select **one** box per row.

Price at Nearby Store	less than the price at Store A	equal to the price at Store A	greater than the price at Store A
0.5 lb for \$1.78	☐	☐	☐
1.3 lbs for \$4.94	☐	☐	☐
2.5 lbs for \$8.70	☐	☐	☐

7. Which expression is equivalent to $-\frac{3}{5}\left(7 - 3\frac{1}{3}\right)$?

- Ⓐ $\left(-\frac{3}{5}\right)(-7) + \left(-\frac{3}{5}\right)\left(-3\frac{1}{3}\right)$
- Ⓑ $-\left(-\frac{3}{5}\right)(7) - \left(-\frac{3}{5}\right)\left(3\frac{1}{3}\right)$
- Ⓒ $-\left(-\frac{3}{5}\right)(7) - \left(-\frac{3}{5}\right)\left(-3\frac{1}{3}\right)$
- Ⓓ $\left(-\frac{3}{5}\right)(7) + \left(-\frac{3}{5}\right)\left(-3\frac{1}{3}\right)$

8. A person paid a total of \$38.50, before tax, for flowers and dirt at a store. The price of the flowers was \$4.50 per container. The price of the dirt was \$7.00 per bag.

If the customer bought one bag of dirt, how many containers of flowers did the customer buy?

- (A) 5
- (B) 7
- (C) 9
- (D) 10

9. Which expression is equivalent to $-\frac{1}{5}(y - 3) + 5 + \frac{3}{10}y$?

Select **all** expressions that are equivalent.

- (A) $-\frac{1}{5}y + 5\frac{3}{5} + \frac{3}{10}y$
- (B) $-\frac{1}{5}y + 4\frac{2}{5} + \frac{3}{10}y$
- (C) $\frac{1}{10}y + 5\frac{3}{5}$
- (D) $\frac{1}{2}y + 4\frac{2}{5}$
- (E) $\frac{1}{10}y + 2$

10. A student spent \$18 for a haircut this week.

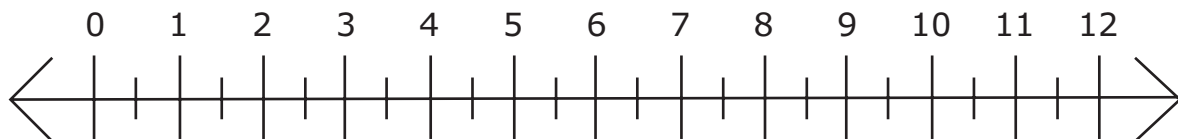
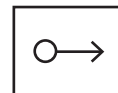
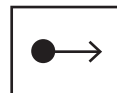
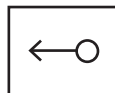
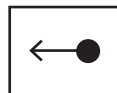
Which event, when combined with the student spending \$18, will result in the student having \$0 remaining?

- ☐ A The student earned \$18 for cleaning out his neighbor's garage.
- ☐ B The student paid his little brother \$18 for doing his chores.
- ☐ C The student purchased a gift for his mom for \$18.
- ☐ D The student gave \$18 to a friend.

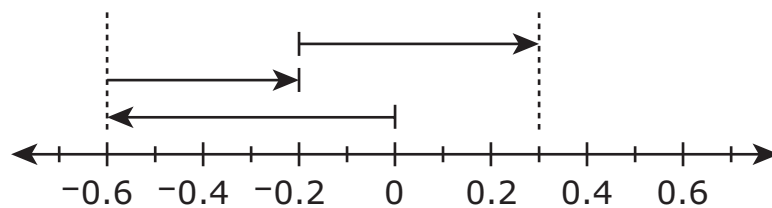
11. A student wants to walk 30 or more total miles this month and has already walked 18 of the 30 total miles. The student walks at a constant speed of 3 miles per hour.

Graph n , all the possible values for the number of additional hours it will take the student to walk at least 30 total miles this month.

Select a ray and then graph the ray so that it begins at the correct place on the number line.



12. The number line model can be used to represent an expression.



Which expression represents the number line model?

- ☐ A $-0.6 + -0.4 + 0.5$
- ☐ B $-0.6 + 0.4 + 0.5$
- ☐ C $-0.6 - 0.3$
- ☐ D $0.3 - -0.6$

This is the end of Item Set 1 Section 1.

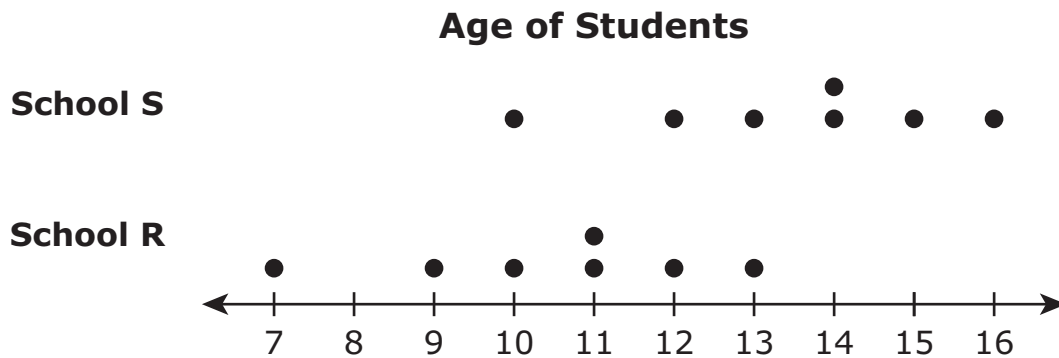


ITEM SET 1 - SECTION 2 (Calculator)

You may use a calculator for Item Set 1 - Section 2.



13. Students from two schools attend a game tournament. Data about the age of the students from each school are shown in the line plot.



The mean age, in years, of the students from School R is 10.4. The mean age, in years, of the students from School S is 13.4. The mean absolute deviation for each group of students is about 1.5.

Circle the answer options to correctly complete the sentence.

The difference between the mean ages is _____ ,

- 3

8.9

11.9

which is about _____ times the mean absolute deviation

- 0.5

2

for either school.

This is the end of Item Set 1 Section 2.



ITEM SET 2 (Calculator)

You may use a calculator for Item Set 2.



1. Part A

Consider the expression $\frac{1}{2}x + 3 + \frac{1}{3}x - 4$.

Write an equivalent expression with exactly two terms.

Enter your expression in the space provided. Enter **only** your expression.

Part B

A student states that the expressions $\frac{1}{2}(x + 3) + \frac{1}{3}(x - 4)$ and $\frac{1}{2}x + 3 + \frac{1}{3}x - 4$ are equivalent.

- Explain why the student's reasoning is incorrect.
- Create an expression, with two terms, that is equivalent to $\frac{1}{2}(x + 3) + \frac{1}{3}(x - 4)$. Show your work or explain your reasoning.

Enter your explanations or work and your answer in the space provided.



2. A person can play $\frac{1}{6}$ of a song in $\frac{1}{3}$ of a minute.

How many minutes does it take the person to play the whole song once at this rate?

(A) 3

(B) 2

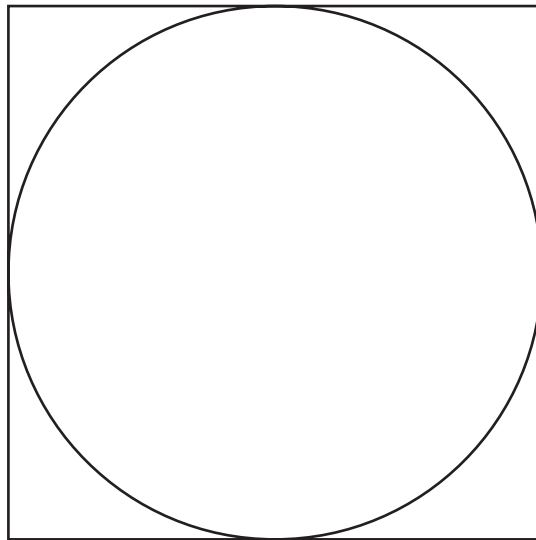
(C) $\frac{2}{3}$

(D) $\frac{1}{2}$



Use the information provided to answer Part A and Part B for question 3.

A circle is inscribed on a square piece of paper. The radius of the circle is 5 centimeters.



3. Part A

What is the circumference, in centimeters, of the circle?

Enter your answer in the space provided. Enter **only** your answer.

_____ **centimeters**

Part B

The circle is cut out and removed from the square. How much paper, in square centimeters, remains after the circle is removed?

- (A) $25\pi - 10$
- (B) $10\pi - 25$
- (C) $100 - (10\pi)$
- (D) $100 - (25\pi)$



Use the information provided to answer Part A through Part D for question 4.

A 50-pound bag of dog food contains 200 cups of food.

- A large dog eats 14 cups of food in 4 days.
- A small dog eats 9 cups of food in 6 days.

The daily serving size of dog food for each dog is constant.

4. Part A

Both dogs are fed from a new 50-pound bag of dog food.

What fraction of the new bag remains after 30 days?

(A) $\frac{1}{4}$

(B) $\frac{19}{40}$

(C) $\frac{3}{4}$

(D) $\frac{31}{40}$

Part B

How many complete daily servings for the large dog are in a 28-pound bag of dog food?

Enter your answer in the box.

⊖					
•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9



Part C

The owner uses a scoop to pour the food into each dog's bowl. A 1-day serving of food for the large dog requires 7 scoops of dog food.

Based on the ratio of 1-day servings for each dog, how many scoops are needed to serve the small dog a 1-day serving of food?

Enter your answer in the space provided. Enter **only** your answer.

Small dog's 1-day serving: _____ **scoops**

Part D

A medium dog eats 11 cups of dog food in 4 days.

How many more cups of food does the large dog eat compared to the medium dog in 90 days?

- ☐ A 270
- ☐ B 180
- ☐ C 112.5
- ☐ D 67.5



5. A researcher wants to predict the number of deer at a park based on the size of the park. The table shows data collected about the size, in acres, and the current number of deer at several parks.

Park Data

Park Name	Size (acres)	Number of Deer
Johnson	1,980	195
Kennedy	1,590	162
Lincoln	1,720	168
Roosevelt	2,050	201
Taft	2,060	198

- Use the data to find an estimated rate of deer per acre for any park. Explain how you determined your estimate.
- Create an equation to model the estimated number of deer, d , that exist in n acres of a park.

Enter your answer, your explanation, and your equation in the space provided.



Use the information provided to answer Part A and Part B for question 6.

A reporter from the school paper asks a random sample of students about their favorite school food. The results are shown in the table. There are 441 students in total at the school.

Favorite School Food

Favorite School Food	Number of Students
spaghetti	12
salad	15
hamburger	11
pizza	20
other	5

6. Part A

Based on the sample, approximately how many students in the entire school population prefer spaghetti as their favorite school food?

- (A) 12
- (B) 63
- (C) 84
- (D) 105

Part B

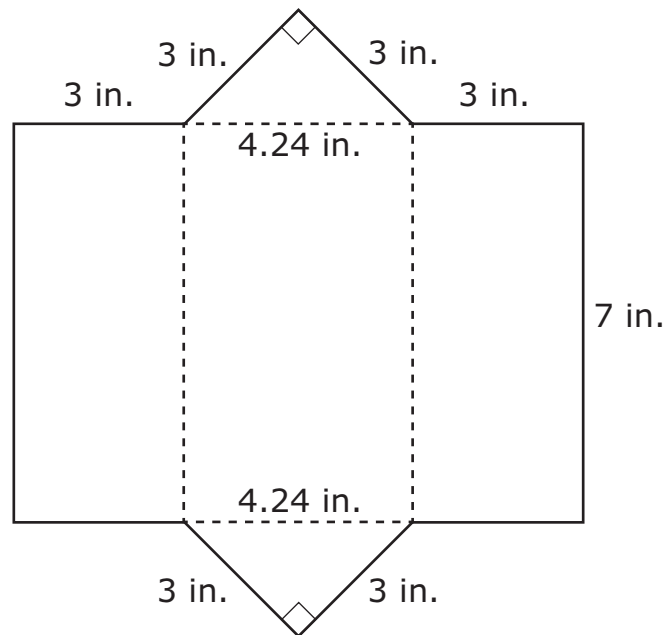
Based on the sample, which inference can be made?

- (A) The number of students who like spaghetti is approximately the same as the number who like hamburgers.
- (B) A small number of students bring their own lunch to school.
- (C) Most students eat salad or pizza every day.
- (D) Almost the whole school prefers pizza.



7. Part A

A person wants to make a pencil case in the shape of a triangular prism. He traces the outline using the pattern shown.





- Show or explain the steps needed to calculate the amount of material, in square inches, needed to make 1 pencil case.
- What is the amount of material, in square inches, needed to make 1 pencil case?
- Create an expression that can be used to determine the amount of material, in square inches, to make n pencil cases.

Enter your explanations, your work, and your answer in the space provided.



Part B

The material for the pencil case is sold in square-shaped pieces that are 1 foot long. Each piece costs \$5.

- Find the area, **in square inches**, of one piece of material.
- What is the cost per square inch of the material? Show or explain your work.

Enter your answer and your work or explanation in the space provided.

This is the end of Item Set 2.



ITEM SET 3 (Calculator)

You may use a calculator for Item Set 3.



1. A turtle travels $7\frac{7}{8}$ miles in $\frac{3}{7}$ of a week. At this rate, how many miles per week will the turtle travel?

Enter your answer in the space provided. Enter **only** your answer.



2. A teacher creates a grading system for a 5-week period.

- There is a weekly quiz worth 18 points.
- There is a weekly homework assignment worth a fixed number of points.
- The total points for all homework assignments during the 5-week period is 170 points.

The equation $5(x + 18) = 170$ represents this situation.

- Show or explain each step to solve for x .
- State the value of x .
- Explain what the solution for x represents in this situation.

Enter your explanations and your answer in the space provided.



3. A teacher has an equal number of pink, blue, green, yellow, and purple erasers in a box. She will randomly distribute 1 eraser from the box to each student.

What is the probability that the first eraser that will be distributed is a color other than yellow?

- (A) $\frac{1}{5}$
- (B) $\frac{1}{4}$
- (C) $\frac{3}{4}$
- (D) $\frac{4}{5}$



4. A person is planning an event and wants to order chicken and beans from a store.

- The store charges \$9.49 per pound of chicken and \$5.75 for one quart of beans.
- The person has \$70 to purchase the chicken and the beans.

Let p represent the number of pounds of chicken.

- Create an equation that can determine the maximum amount of chicken, in pounds, that can be purchased, along with one quart of beans, using the \$70. Do not include tax.
- Solve the equation to determine the maximum amount of chicken that can be purchased. Round your final answer to the nearest quarter pound. Show your work.

Enter your equation, your solution, and your work in the space provided.

Equation: _____

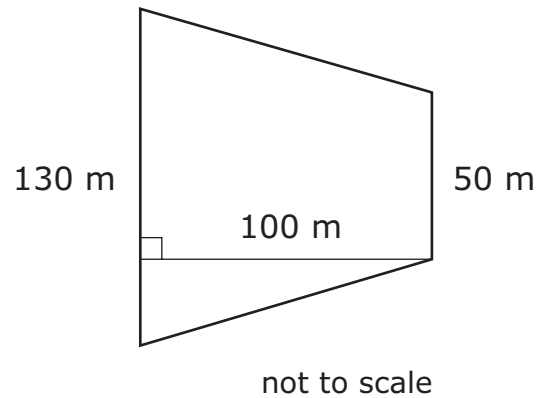
Maximum amount of chicken: _____ **pounds**

Your Work:



5. Part A

A piece of land is in the shape of a trapezoid¹, with parallel sides that are 50 meters and 130 meters long. The distance between the parallel sides is 100 meters, as shown in the diagram.



How large, in square meters, is the piece of land?

Enter your answer in the box.

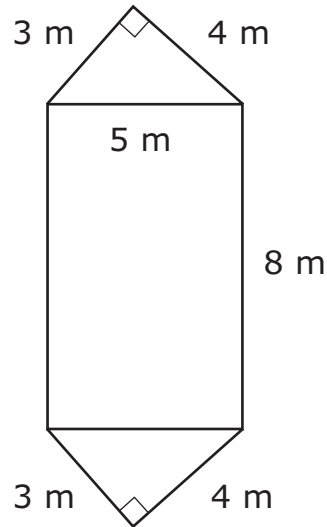
⊖					
•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

¹trapezoid—a quadrilateral with at least one pair of parallel sides



Part B

There is a large picnic area on the property. It is made up of a rectangle, with dimensions of 5 meters by 8 meters, and two right triangles, whose legs are 3 meters and 4 meters long, as shown in the diagram.



How large, in square meters, is the picnic area?

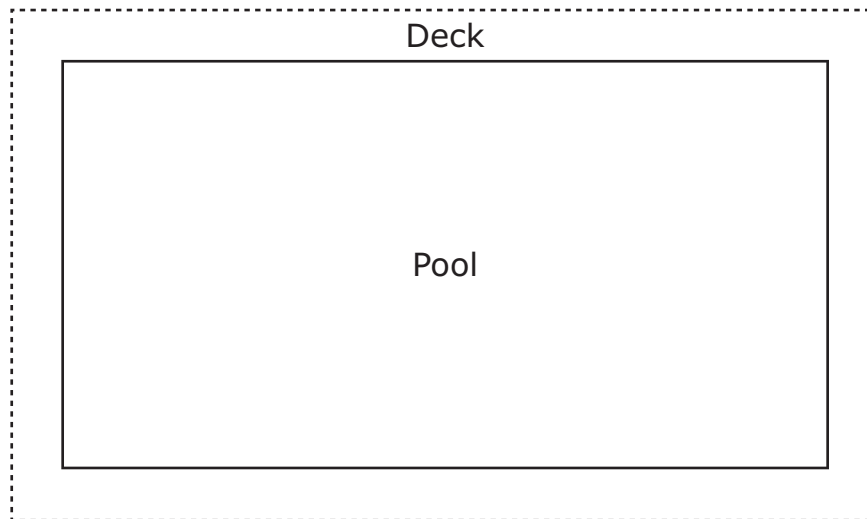
Enter your answer in the box.

⊖					
•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9



Use the information provided to answer Part A and Part B for question 6.

A person is building a deck around a rectangular pool, as shown in the diagram.



- The width, W , of the pool is 8 feet, and the length, L , of the pool is 15 feet.
- The expression $2(W + 12) + 2(L + 12)$ can be used to find the total perimeter of the deck, in feet.
- The person uses a different expression, $2W + 2L + 48$, to find the perimeter of the deck.

6. Part A

How many terms are in the expression $2W + 2L + 48$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5



Part B

- Explain whether the two expressions for finding the perimeter of the deck are equivalent.
- The person claims the perimeter of the deck will be twice the perimeter of the pool. Explain whether the person's claim is correct or incorrect.
- State the perimeter of the pool and the perimeter of the deck.

Enter your explanations and your answers in the space provided.

This is the end of Item Set 3.



Colorado Measures of Academic Success



Grade 7 Math

Answer Key with Scoring Rubrics, Sample Responses & Annotations

Practice Resource for Students

ANSWER KEY: ITEM SET 1

Item Set 1 – Question 1 (Selected Response)

Which expression is equivalent to $\frac{1}{2} - \frac{3}{5}$?

☐ A. $-\frac{3}{5} + \left(-\frac{1}{2}\right)$

☐ B. $-\frac{1}{2} + \left(\frac{3}{5}\right)$

☒ C. $\frac{1}{2} + \left(-\frac{3}{5}\right)$

☐ D. $\frac{3}{5} + \left(\frac{1}{2}\right)$

Item Information

- Answer – C
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.NS.A.1.c
 - Demonstrate subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in real-world contexts.
- Evidence Statement – 7.NS.1c-1
 - Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram. c. Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Apply this principle in real-world contexts. i) Pool should contain tasks with and without context. ii) Contextual tasks might, for example, require students to create or identify a situation described by a specific equation of the general form $p - q = p + (-q)$ such as $3 - 5 = 3 + (-5)$. iii) Non-contextual tasks are not computation tasks but rather require students to demonstrate conceptual understanding, for example by identifying a sum that is equivalent to a given difference. For example, given the difference $-1/3 - (1/5 + 5/8)$, the student might be asked to recognize the equivalent expression $-1/3 + -(1/5 + 5/8)$. iv) Tasks are not limited to integers.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.536

Item Set 1 – Question 2 (TEI Equation Editor)

In 1 year, the water level of a lake changes by $-\frac{3}{8}$ inch.

If the water level of the lake continues to change at this rate for 7 years, how many inches will the water level of the lake have changed?

Enter your answer in the space provided. Enter **only** your answer.

$-2\frac{5}{8}$

Item Information

- Answer – See Image
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.NS.A.3
 - Solve real-world and mathematical problems involving the four operations with rational numbers. (Computations with rational numbers extend the rules for manipulating fractions to complex fractions.)
- Evidence Statement – 7.NS.3
 - Solve real-world and mathematical problems involving the four operations with rational numbers. i) Tasks are one-step word problems. ii) Tasks sample equally between addition/subtraction and multiplication/division. iii) Tasks involve at least one negative number. iv) Tasks are not limited to integers.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.324

Item Set 1 – Question 3 (Selected Response)

In one hour, Earth travels approximately 67,000 miles in its orbit around the Sun.

Which equation represents the relationship between the total number of miles, m , that Earth travels in h hours?

- ☐ A. $67,000 + m = h$
- ☐ B. $67,000 + h = m$
- ☐ C. $67,000m = h$
- ☒ D. $67,000h = m$

Item Information

- Answer – D
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.RP.A.2.c
 - Represent proportional relationships by equations. For example, if total cost t is proportional to the number n of items purchased at a constant price p , the relationship between the total cost and the number of items can be expressed as $t = pn$.
- Evidence Statement – 7.RP.2c
 - Recognize and represent proportional relationships between quantities: c. Represent proportional relationships by equations. For example, if total cost t is proportional to the number n of items purchased at a constant price p , the relationship between the total cost and the number of items can be expressed as $t = pn$. i) Tasks have a context. ii) Tasks will include proportional relationships that only involve positive numbers.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.221

Item Set 1 – Question 4 (TEI Drag and Drop)

Create an expression that has the same value as $(6x - 4) + (x + 5)$.

Drag and drop the numbers into the expression. Each number may be used once, more than once, or not at all.

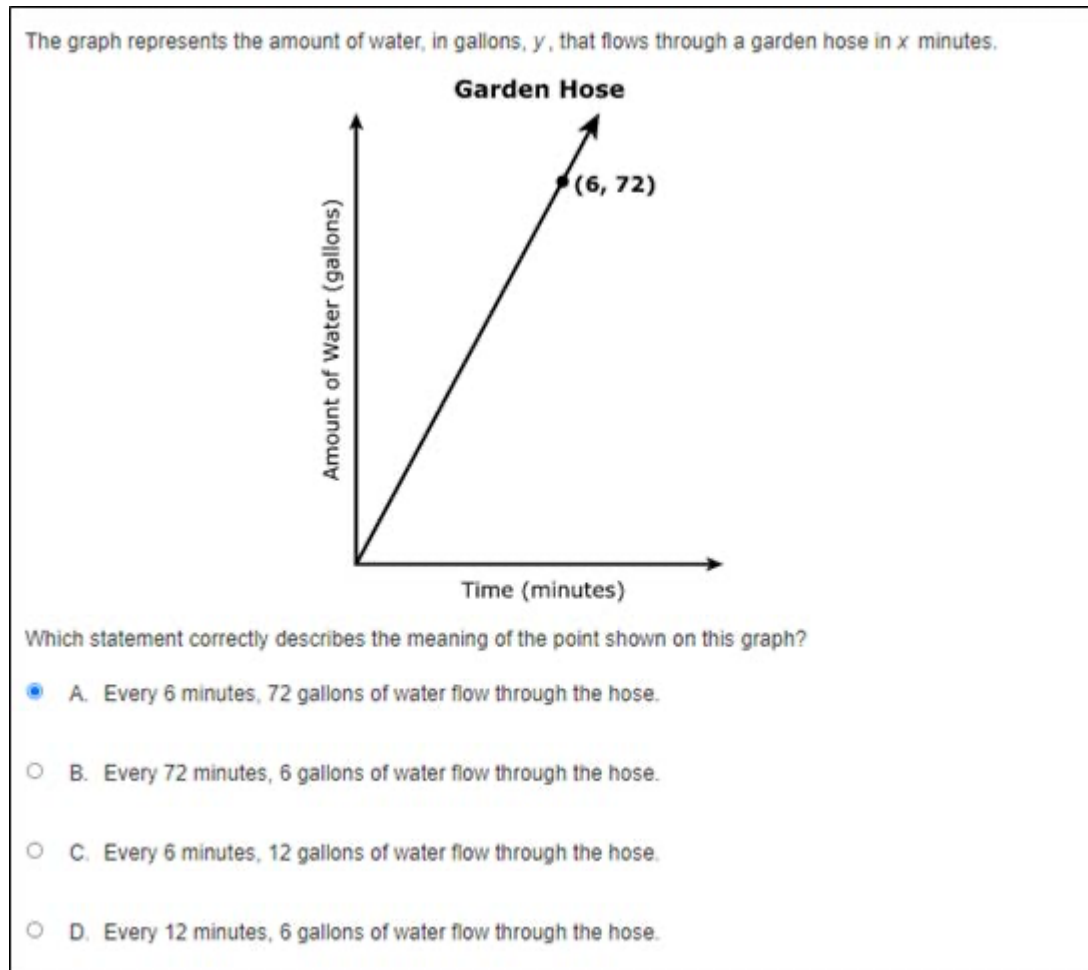
1 2 4 5 6 7 9 11

7 $x +$ 1

Item Information

- Answer – See Image
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.EE.A.1
 - Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.
- Evidence Statement – 7.EE.1
 - Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients. i) Tasks are not limited to integer coefficients. ii) Tasks may involve issues of strategy, e.g., by providing a factored expression such as $y(3+x+k)$ and a fully expanded expression $3y + xy + ky$, and requiring students to produce or identify a new expression equivalent to both (such as $y(3+x) + yk$).
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.31

Item Set 1 – Question 5 (Selected Response)



Item Information

- Answer – A
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.RP.A.2.d
 - Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points $(0, 0)$ and $(1, r)$ where r is the unit rate.
- Evidence Statement – 7.RP.2d
 - Recognize and represent proportional relationships between quantities. d. Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points $(0, 0)$ and $(1, r)$ where r is the unit rate. i) Tasks require students to interpret a point (x, y) on the graph of a proportional relationship in terms of the situation, with special attention to the points $(0, 0)$ and $(1, r)$ where r is the unit rate. For the explanation aspect of standard 7.RP.2d, see 7.C.6.1. ii) Tasks will include proportional relationships that only involve positive numbers.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.638

Item Set 1 – Question 6 (TEI Multiple Select)

Store A has meat on sale for $1\frac{1}{2}$ lbs for \$5.34. The table shows the prices for meat at three nearby stores.

Select the boxes to identify how the price of the meat at each nearby store compares to the price of the meat at Store A.

Price at Nearby Store	less than the price at Store A	equal to the price at Store A	greater than the price at Store A
0.5 lb for \$1.78	☐	☒	☐
1.3 lbs for \$4.94	☐	☐	☒
2.5 lbs for \$8.70	☒	☐	☐

Item Information

- Answer – See Image
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.RP.A.2.b
 - Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.
- Evidence Statement – 7.RP.2b
 - Recognize and represent proportional relationships between quantities: b. Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships. i) Pool should contain tasks with and without context. ii) Tasks sample equally across the listed representations (graphs, equations, diagrams, and verbal descriptions). iii) Tasks will include proportional relationships that only involve positive numbers.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.2

Item Set 1 – Question 7 (Selected Response)

Which expression is equivalent to $-\frac{3}{5} \left(7 - 3\frac{1}{3} \right)$?

- ☐ A. $\left(-\frac{3}{5}\right)(-7) + \left(-\frac{3}{5}\right)\left(-3\frac{1}{3}\right)$
- ☐ B. $-\left(-\frac{3}{5}\right)(7) - \left(-\frac{3}{5}\right)\left(3\frac{1}{3}\right)$
- ☐ C. $-\left(-\frac{3}{5}\right)(7) - \left(-\frac{3}{5}\right)\left(-3\frac{1}{3}\right)$
- ☒ D. $\left(-\frac{3}{5}\right)(7) + \left(-\frac{3}{5}\right)\left(-3\frac{1}{3}\right)$

Item Information

- Answer – D
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.NS.A.2.a
 - Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts.
- Evidence Statement – 7.NS.2a-1
 - Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers. a. Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. i) Tasks do not have a context. ii) Tasks are not computation tasks but rather require students to demonstrate conceptual understanding, for example by providing students with a numerical expression and requiring students to produce or recognize an equivalent expression using properties of operations, particularly the distributive property. For example, given the expression $(-3)(6 + -4 + -3)$, the student might be asked to recognize that the given expression is equivalent to $(-3)(6 + -4) + (-3)(-3)$.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.377

Item Set 1 – Question 8 (Selected Response)

A person paid a total of \$38.50, before tax, for flowers and dirt at a store. The price of the flowers was \$4.50 per container. The price of the dirt was \$7.00 per bag.

If the customer bought one bag of dirt, how many containers of flowers did the customer buy?

- ☐ A. 5
- ☒ B. 7
- ☐ C. 9
- ☐ D. 10

Item Information

- Answer – B
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.EE.B.4.a
 - Solve word problems leading to equations of the form $px (+/-) q = r$ and $p(x (+/-) q) = r$, where p , q and r are specific rational numbers. Solve equations of these forms fluently. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. For example, the perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?
- Evidence Statement – 7.EE.4a-1
 - Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities. a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p , q , and r are specific rational numbers. i) Comparison of an algebraic solution to an arithmetic solution is not assessed here; for this aspect of 7.EE.4a, see 7.C.5.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.663

Item Set 1 – Question 9 (Multiple Select)

Which expression is equivalent to $-\frac{1}{5}(y - 3) + 5 + \frac{3}{10}y$?

Select **all** expressions that are equivalent.

☒ A. $-\frac{1}{5}y + 5\frac{3}{5} + \frac{3}{10}y$

☐ B. $-\frac{1}{5}y + 4\frac{2}{5} + \frac{3}{10}y$

☒ C. $\frac{1}{10}y + 5\frac{3}{5}$

☐ D. $\frac{1}{2}y + 4\frac{2}{5}$

☐ E. $\frac{1}{10}y + 2$

Item Information

- Answer – A, C
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.EE.A.1
 - Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.
- Evidence Statement – 7.EE.1
 - Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients. i) Tasks are not limited to integer coefficients. ii) Tasks may involve issues of strategy, e.g., by providing a factored expression such as $y(3+x+k)$ and a fully expanded expression $3y + xy + ky$, and requiring students to produce or identify a new expression equivalent to both (such as $y(3+x) + yk$).
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.271

Item Set 1 – Question 10 (Selected Response)

A student spent \$18 for a haircut this week.

Which event, when combined with the student spending \$18, will result in the student having \$0 remaining?

- ☒ A. The student earned \$18 for cleaning out his neighbor's garage.
- ☐ B. The student paid his little brother \$18 for doing his chores.
- ☐ C. The student purchased a gift for his mom for \$18.
- ☐ D. The student gave \$18 to a friend.

Item Information

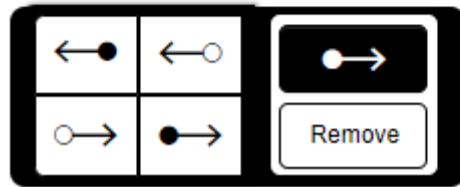
- Answer – A
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.NS.A.1.a
 - Describe situations in which opposite quantities combine to make 0. For example, a hydrogen atom has 0 charge because its two constituents are oppositely charged.
- Evidence Statement – 7.NS.1a
 - Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram. a. Describe situations in which opposite quantities combine to make 0. For example, a hydrogen atom has 0 charge because its two constituents are oppositely charged. i) Tasks require students to recognize or identify situations of the kind described in standard 7.NS.1a.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.62

Item Set 1 – Question 11 (TEI Number Line)

A student wants to walk 30 or more total miles this month and has already walked 18 of the 30 total miles. The student walks at a constant speed of 3 miles per hour.

Graph n , all the possible values for the number of additional hours it will take the student to walk at least 30 total miles this month.

Select a ray. Drag the point on the ray to the appropriate location on the number line.

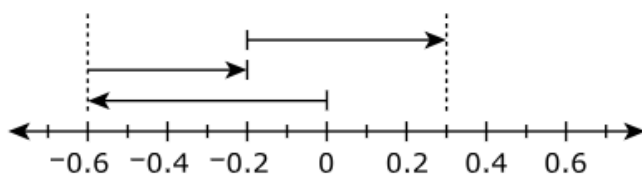


Item Information

- Answer – See Image
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.EE.B.4.b
 - Solve word problems leading to inequalities of the form $px (+/-) q > r$, $px (+/-) q \geq r$, $px (+/-) q < r$, or $px (+/-) q \leq r$, where p , q and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem. For example: As a salesperson, you are paid \$50 per week plus \$3 per sale. This week you want your pay to be at least \$100. Write an inequality for the number of sales you need to make and describe the solutions.
- Evidence Statement – 7.EE.4b
 - Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities. b. Solve word problems leading to inequalities of the form $px + q > r$ or $px + q < r$, where p , q and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem. For example: As a salesperson, you are paid \$50 per week plus \$3 per sale. This week you want your pay to be at least \$100. Write an inequality for the number of sales you need to make, and describe the solutions. i) Tasks may involve $<$, $>$, $\leq$ or $\geq$.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – This item was developed as a practice item and does not have a P-value.

Item Set 1 – Question 12 (Selected Response)

The number line model can be used to represent an expression.



Which expression represents the number line model?

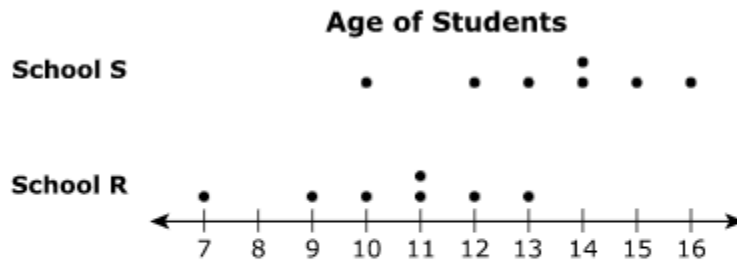
- ☐ A. $-0.6 + -0.4 + 0.5$
- ☒ B. $-0.6 + 0.4 + 0.5$
- ☐ C. $-0.6 - 0.3$
- ☐ D. $0.3 - -0.6$

Item Information

- Answer – B
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.NS.A.1.d
 - Apply properties of operations as strategies to add and subtract rational numbers.
- Evidence Statement – 7.NS.1d
 - Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram. d. Apply properties of operations as strategies to add and subtract rational numbers. i) Tasks do not have a context. ii) Tasks are not limited to integers. iii) Tasks may involve sums and differences of 2 or 3 rational numbers. iv) Tasks require students to represent addition and subtraction on a horizontal or vertical number line or compute a sum or difference, or demonstrate conceptual understanding for example by producing or recognizing an expression equivalent to a given sum or difference. For example, given the sum $-8.1 + 7.4$, the student might be asked to recognize or produce the equivalent expression $-(8.1 - 7.4)$.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.494

Item Set 1 – Question 13 (TEI Inline Choice)

Students from two schools attend a game tournament. Data about the age of the students from each school are shown in the line plot.



The mean age, in years, of the students from School R is 10.4. The mean age, in years, of the students from School S is 13.4. The mean absolute deviation for each group of students is about 1.5.

Select from the drop-down menus to correctly complete the sentence.

The difference between the mean ages is , which is about times the mean absolute deviation for either school.

Item Information

- Answer – See Image
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.SP.B.3
 - Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, measuring the difference between the centers by expressing it as a multiple of a measure of variability. For example, the mean height of players on the basketball team is 10 cm greater than the mean height of players on the soccer team, about twice the variability (mean absolute deviation) on either team; on a dot plot, the separation between the two distributions of heights is noticeable.
- Evidence Statement – 7.SP.3
 - Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, measuring the difference between the centers by expressing it as a multiple of a measure of variability. For example, the mean height of players on the basketball team is 10cm greater than the mean height of players on the soccer team, about twice the variability (mean absolute deviation) on either team; on a dot plot, the separation between the two distributions of heights is noticeable. i) Tasks may use mean absolute deviation, range, or interquartile range as a measure of variability. ii) Tasks may include pairing concepts of mean absolute deviation and mean or median and interquartile range.
- Subclaim – B – Supporting Content
 - The student solves problems involving the Additional and Supporting Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.383

ANSWER KEY: ITEM SET 2

Item Set 2 – Question 1 (TEI Equation Editor, Constructed Response)

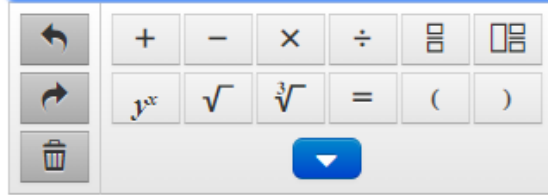
Part A

Consider the expression $\frac{1}{2}x + 3 + \frac{1}{3}x - 4$.

Write an equivalent expression with exactly two terms.

Enter your expression in the space provided. Enter **only** your expression.

$$\frac{5}{6}x - 1$$



Part B

A student states that the expressions $\frac{1}{2}(x + 3) + \frac{1}{3}(x - 4)$ and $\frac{1}{2}x + 3 + \frac{1}{3}x - 4$ are equivalent.

- Explain why the student's reasoning is incorrect.
- Create an expression, with two terms, that is equivalent to $\frac{1}{2}(x + 3) + \frac{1}{3}(x - 4)$. Show your work or explain your reasoning.

Enter your explanations or work and your answer in the space provided.

Item Information

- Answer – See Scoring Rubric and Sample Student Responses
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.EE.A.1
 - Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.
- Evidence Statement – 7.C.1-2
 - Base explanations/reasoning on the properties of operations. Content Scope: Knowledge and skills articulated in 7.EE.1. i) Tasks should not require students to identify or name properties.
- Subclaim – C – Expressing Mathematical Reasoning
 - The student expresses grade/course-level appropriate mathematical reasoning by constructing viable arguments, critiquing the reasoning of others, and/or attending to precision when making mathematical statements.
- Score Point Distribution
 - 2.9% of students earned 4 points.
 - 3.9% of students earned 3 points.
 - 9.5% of students earned 2 points.
 - 17.4% of students earned 1 point.
 - 66.3% of students earned 0 points.

Scoring Rubric – Part A

Points	Attributes
1	Computation Component: Student response: $\frac{5}{6}x - 1$ or an equivalent expression with exactly two terms. Variable substitution is not allowed. $\frac{1}{2}x + 3 + \frac{1}{3}x - 4$ $\frac{3}{6}x + 3 + \frac{2}{6}x - 4$ $\frac{5}{6}x - 1$
0	Student response is incorrect or irrelevant.

Scoring Rubric – Part B

Points	Attributes
3	Student response includes each of the following 3 elements. Reasoning component: Valid explanation for why the student's reasoning is incorrect Computation component: Correct expression, with two terms, that is equivalent to $\frac{1}{2} + (x + 3) + \frac{1}{3}(x - 4)$ Reasoning component: Valid explanation or work for how the equivalent expression was determined
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Sample Student Response:	Sample Solution 1: The student's reasoning is incorrect because instead of distributing both the $\frac{1}{2}$ and $\frac{1}{3}$ to both terms that make up the expression in the parenthesis they just distributed and multiplied the x's by $\frac{1}{2}$ and $\frac{1}{3}$ and not the 3 and -4. A correct equivalent expression is $\frac{5}{6}x + \frac{1}{6}$. You can find this expression by distributing and multiplying the variable that is outside the parenthesis to both terms inside the parenthesis and get these terms: $1\frac{1}{2} + \frac{1}{2}x + \frac{1}{3}x - 1\frac{1}{3}$. You can then make a common denominator out of all the fractions with 6, then combine and simplify like terms and get the equivalent expression $\frac{5}{6}x + \frac{1}{6}$.

Annotation for	Solution 1, Score Point 3
Sample Student	The response receives full credit. It includes each of the 3 required elements.
Response:	Reasoning Component: Student Response: The student's reasoning is incorrect because instead of distributing both the $\frac{1}{2}$ and $\frac{1}{3}$ to both terms that make up the expression in the parenthesis they just distributed and multiplied the x's by $\frac{1}{2}$ and $\frac{1}{3}$ and not the 3 and -4. Rationale for Score: A valid explanation is provided for why the student's reasoning is incorrect. The student explains that the fraction outside the parenthesis was only distributed to the first variable, x, and not to the other number inside the parenthesis (instead of distributing both the $\frac{1}{2}$ and $\frac{1}{3}$ to both terms that make up the expression in the parenthesis they just distributed and multiplied the x's by $\frac{1}{2}$ and $\frac{1}{3}$ and not the 3 and -4). Computation Component: Student Response: A correct equivalent expression is $\frac{5}{6}x + \frac{1}{6}$ Rationale for score: A correct equivalent expression with two terms is provided ($\frac{5}{6}x + \frac{1}{6}$). Reasoning Component: Student Response: You can find this expression by distributing and multiplying the variable that is outside the parenthesis to both terms inside the parenthesis and get these terms: $1\frac{1}{2} + \frac{1}{2}x + \frac{1}{3}x - 1\frac{1}{3}$. You can then make a common denominator out of all the fractions with 6, then combine and simplify like terms and get the equivalent expression $\frac{5}{6}x + \frac{1}{6}$. Rationale for score: A valid explanation is provided for how the equivalent expression was determined. The student explains that the fraction outside the parenthesis must be correctly distributed and multiplied to both terms inside the parenthesis, then the expression can be simplified to two terms (You can find this expression by distributing and multiplying the variable that is outside the parenthesis to both terms inside the parenthesis and get these terms: $1\frac{1}{2} + \frac{1}{2}x + \frac{1}{3}x - 1\frac{1}{3}$. You can then make a common denominator out of all the fractions with 6, then combine and simplify like terms and get the equivalent expression $\frac{5}{6}x + \frac{1}{6}$). Note: Sample student responses are not representative of all correct answers for an item and are only provided as a guide to assist teachers with scoring.

Item Set 2 – Question 2 (Selected Response)

A person can play $\frac{1}{6}$ of a song in $\frac{1}{3}$ of a minute.

How many minutes does it take the person to play the whole song once at this rate?

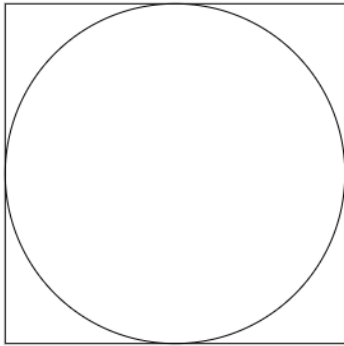
- ☐ A. 3
- ☒ B. 2
- ☐ C. $\frac{2}{3}$
- ☐ D. $\frac{1}{2}$

Item Information

- Answer – B
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.RP.A.1
 - Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities measured in like or different units. For example, if a person walks $\frac{1}{2}$ mile in each $\frac{1}{4}$ hour, compute the unit rate as the complex fraction $\frac{1/2}{1/4}$ miles per hour, equivalently 2 miles per hour.
- Evidence Statement – 7.RP.1
 - Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. For example, if a person walks $\frac{1}{2}$ mile in each $\frac{1}{4}$ hour, compute the unit rate as the complex fraction $\frac{1/2}{1/4}$ miles per hour, equivalently 2 miles per hour. i) Tasks have a real-world context. ii) An example of like units would be cups of butter to cups of sugar; an example of different units would be miles to hours. Unit conversion is not assessed here.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.62

Item Set 2 – Question 3 (TEI Equation Editor, Selected Response)

A circle is inscribed on a square piece of paper. The radius of the circle is 5 centimeters.



Part A

What is the circumference, in centimeters, of the circle?

Enter your answer in the space provided. Enter **only** your answer.

31.40 centimeters

	+	-	×	÷	$\frac{\Box}{\Box}$	$\frac{\Box}{\Box}$
	$x^{\Box}$	$\sqrt{\Box}$	$\sqrt[3]{\Box}$	=	(	)

Part B

The circle is cut out and removed from the square. How much paper, in square centimeters, remains after the circle is removed?

- ☐ A. $25\pi - 10$
- ☐ B. $10\pi - 25$
- ☐ C. $100 - (10\pi)$
- ☒ D. $100 - (25\pi)$

Item Information

- Part A Answer – See Rubric
- Part B Answer – D
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.G.B.4
 - State the formulas for the area and circumference of a circle and use them to solve problems; give an informal derivation of the relationship between the circumference and area of a circle.
- Evidence Statement – 7.G.4-1
 - Know the formulas for the area and circumference of a circle and use them to solve problems. i) Pool should contain tasks with and without context. ii) Tasks may require answers to be written in terms of π .
- Subclaim – B – Supporting Content
 - The student solves problems involving the Additional and Supporting Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.242

Scoring Rubric – Part A

Points	Attributes
1	Student response is a fraction or decimal value within the range of possible values: 31.40 to 31.43 inclusive or exactly 10π .
0	Student response is incorrect or irrelevant.

Item Set 2 – Question 4 (Selected Response, Fill in the Blank, Equation Editor)

A 50-pound bag of dog food contains 200 cups of food.

- A large dog eats 14 cups of food in 4 days.
- A small dog eats 9 cups of food in 6 days.

The daily serving size of dog food for each dog is constant.

Part A

Both dogs are fed from a new 50-pound bag of dog food.

What fraction of the new bag remains after 30 days?

- ☒ A. $\frac{1}{4}$
- ☐ B. $\frac{19}{40}$
- ☐ C. $\frac{3}{4}$
- ☐ D. $\frac{31}{40}$

Part B

How many complete daily servings for the large dog are in a 28-pound bag of dog food?

Enter your answer in the box.

32

Part C

The owner uses a scoop to pour the food into each dog's bowl. A 1-day serving of food for the large dog requires 7 scoops of dog food.

Based on the ratio of 1-day servings for each dog, how many scoops are needed to serve the small dog a 1-day serving of food?

Enter your answer in the space provided. Enter **only** your answer.

Small dog's 1-day serving : 3 scoops

Part D

A medium dog eats 11 cups of dog food in 4 days.

How many more cups of food does the large dog eat compared to the medium dog in 90 days?

- ☐ A. 270
- ☐ B. 180
- ☐ C. 112.5
- ☒ D. 67.5

Item Information

- Part A Answer – A
- Part B Answer – See Image
- Part C Answer – See Image, or equivalent fraction or decimal.
- Part D Answer – D
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.RP.A.3
 - Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.
- Evidence Statement – 7.RP.3-1
 - Use proportional relationships to solve multistep ratio problems. i) Tasks will include proportional relationships that only involve positive numbers.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.292

Item Set 2 – Question 5 (Constructed Response)

A researcher wants to predict the number of deer at a park based on the size of the park. The table shows data collected about the size, in acres, and the current number of deer at several parks.

Park Data		
Park Name	Size (acres)	Number of Deer
Johnson	1,980	195
Kennedy	1,590	162
Lincoln	1,720	168
Roosevelt	2,050	201
Taft	2,060	198

- Use the data to find an estimated rate of deer per acre for any park. Explain how you determined your estimate.
- Create an equation to model the estimated number of deer, d , that exist in n acres of a park.

Enter your answer, your explanation, and your equation in the space provided.

Item Information

- Answer – See Scoring Rubric and Sample Student Responses
- Colorado Academic Standards (CAS) Evidence Outcomes
 - 7.RP.A.2.b – Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.
 - 7.RP.A.2.c – Represent proportional relationships by equations. For example, if total cost t is proportional to the number n of items purchased at a constant price p , the relationship between the total cost and the number of items can be expressed as $t = pn$.
- Evidence Statement – 7.D.3
 - Micro-models: Autonomously apply a technique from pure mathematics to a real-world situation in which the technique yields valuable results even though it is obviously not applicable in a strict mathematical sense (e.g., profitably applying proportional relationships to a phenomenon that is obviously nonlinear or statistical in nature). Content Scope: Knowledge and skills articulated in Type I, Sub-Claim A Evidence Statements. i) Tasks may have scaffolding, if necessary, in order to yield a degree of difficulty appropriate to Grade 7.
- Subclaim – D – Modeling and Application
 - The student solves real-world problems with a degree of difficulty appropriate to the grade/course by applying knowledge and skills articulated in the standards for the current grade/course (or for more complex problems, knowledge and skills articulated in the standards for previous grades/courses), engaging particularly in the Modeling practice, and where helpful making sense of problems and persevering to solve them (MP. 1), reasoning abstractly and quantitatively (MP. 2), using appropriate tools strategically (MP.5), looking for and making use of structure (MP.7), and/or looking for and expressing regularity in repeated reasoning (MP.8).
- Score Point Distribution
 - 1.8% of students earned 3 points.
 - 9.0% of students earned 2 points.
 - 30.5% of students earned 1 point.
 - 58.7% of students earned 0 points.

Points	Attributes
3	Student response includes the following 3 elements. • Modeling component = 1 point: Reasonably estimated rate of deer per acre. • Modeling Component = 1 point: Valid explanation for how to determine an estimated rate of deer per acre. • Modeling component = 1 point: Valid equation to model the number of estimated deer, d, in n acres of a park. Sample Student Response: To estimate the rate of deer per acre, I found the unit rate for the number of deer per acre for each park by dividing the number of deer by the number of acres. Then, I added up the unit rates for each park and divided by 5 to find the average rate, which is about 0.098. Then, I rounded the 0.098 to 0.1 deer per acre. Using the rate of 0.1 deer per acre, I created equation $d=0.1n$ to model the number of deer, d, in n acres of a park. Note: • Estimates for the rate can vary. • No variable substitution allowed in the equation. If an incorrect rate, from the first prompt, was used in the equation correctly, then 1 point is earned for the equation.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Sample Student Response:	Sample Solution 1: My estimate for Johnson Park is that there is 0.1 deer for every acre of land because 1980 acres can be round to 2,000 and 195 deer can be rounded to 200 and $2000 \div 2000 = 1$, so $200 \div 2000 = 0.1$. $d = 0.1n$
Annotation for Sample Student Response:	Solution 1, Score Point 3 The response receives full credit. It includes each of the 3 required elements. Modeling Component: • Student Response: 0.1 deer for every acre of land ○ Rationale for Score: Valid reasonable estimate of the rate of deer per acre is provided (My estimate for Johnson Park is that there is 0.1 deer for every acre of land). Estimates that are in the range of 0.09 – 0.102 are considered acceptable to earn credit for this element. Modeling Component: • Student Response: 1980 acres can be round to 2,000 and 195 deer can be rounded to 200 and $2000 \div 2000 = 1$, so $200 \div 2000 = 0.1$ ○ Rationale for score: Valid explanation provided for how the estimated rate of deer per acre was determined (1980 acres can be round to 2,000 and 195 deer can be rounded to 200 and $2000 \div 2000 = 1$, so $200 \div 2000 = 0.1$). It is acceptable to use data from one of the five parks to estimate the rate of deer per acre. Modeling Component: • Student Response: $d = 0.1n$ ○ Rationale for score: Valid equation to model the number of estimated deer, d, in n acres is provided ($d = 0.1n$). The provided equation uses the variables as defined in the prompt, along with the estimated rate given in the response. Note: Sample student responses are not representative of all correct answers for an item and are only provided as a guide to assist teachers with scoring.

Part A

Based on the sample, approximately how many students in the entire school population prefer spaghetti as their favorite school food?

- ☐ A. 12
- ☐ B. 63
- ☒ C. 84
- ☐ D. 105

Part B

Based on the sample, which inference can be made?

- ☒ A. The number of students who like spaghetti is approximately the same as the number who like hamburgers.
- ☐ B. A small number of students bring their own lunch to school.
- ☐ C. Most students eat salad or pizza every day.
- ☐ D. Almost the whole school prefers pizza.

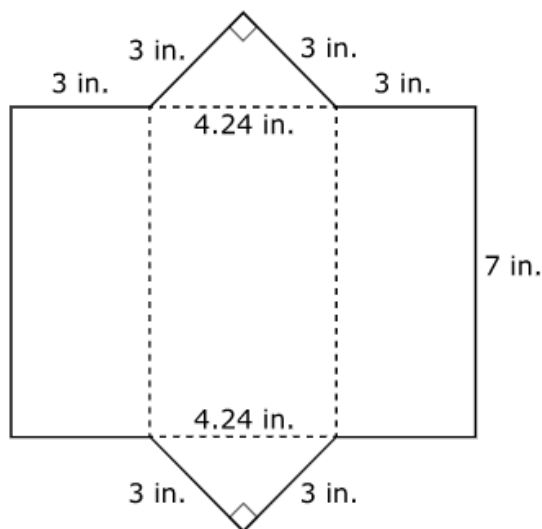
Item Information

- Part A Answer – C
- Part B Answer – A
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.SP.A.2
 - Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions. For example, estimate the mean word length in a book by randomly sampling words from the book; predict the winner of a school election based on randomly sampled survey data. Gauge how far off the estimate or prediction might be.
- Evidence Statement – 7.SP.2
 - Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions. For example, estimate the mean word length in a book by randomly sampling words from the book; predict the winner of a school election based on randomly sampled survey data. Gauge how far off the estimate or prediction might be.
- Subclaim – B – Supporting Content
 - The student solves problems involving the Additional and Supporting Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.326

Item Set 2 – Question 7 (Constructed Response)

Part A

A person wants to make a pencil case in the shape of a triangular prism. He traces the outline using the pattern shown.



- Show or explain the steps needed to calculate the amount of material, in square inches, needed to make 1 pencil case.
- What is the amount of material, in square inches, needed to make 1 pencil case?
- Create an expression that can be used to determine the amount of material, in square inches, to make n pencil cases.

Enter your explanations, your work, and your answer in the space provided.

Part B

The material for the pencil case is sold in square-shaped pieces that are 1 foot long. Each piece costs \$5.

- Find the area, **in square inches**, of one piece of material.
- What is the cost per square inch of the material? Show or explain your work.

Enter your answer and your work or explanation in the space provided.

Item Information

- Answer – See Scoring Rubric and Sample Response
- Colorado Academic Standards (CAS) Evidence Outcomes
 - 6.G.A.4 – Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.
 - 6.G.A.1 – Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles and other shapes; apply these techniques in the context of solving real-world and mathematical problems.
 - 6.RP.A.2 – Apply the concept of a unit rate $a:b$ associated with a ratio $a:b$ with $b \neq 0$, and use rate language in the context of a ratio relationship. For example, "This recipe has a ratio of 3 cups of flour to 4 cups of sugar, so there is $\frac{3}{4}$ cup of flour for each cup of sugar." "We paid \$75 for 15 hamburgers, which is a rate of \$5 per hamburger." (Expectations for unit rates in this grade are limited to non-complex fractions.)
 - 6.RP.A.3b – Solve unit rate problems including those involving unit pricing and constant speed. For example, if it took 7 hours to mow 4 lawns, then at that rate, how many lawns could be mowed in 35 hours? At what rate were lawns being mowed? The ABBI forms test map configuration will need to be updated to supply all 4 evidence outcomes and content will do those updates.
- Evidence Statement – 7.D.2
 - Solve multi-step contextual problems with degree of difficulty appropriate to grade 7, requiring application of knowledge and skills articulated in 6.RP.A, 6.EE.C, and 6.G. i) Tasks may have scaffolding, if necessary, in order to yield a degree of difficulty appropriate to Grade 7.
- Subclaim – D – Modeling and Application
 - The student solves real-world problems with a degree of difficulty appropriate to the grade/course by applying knowledge and skills articulated in the standards for the current grade/course (or for more complex problems, knowledge and skills articulated in the standards for previous grades/courses), engaging particularly in the Modeling practice, and where helpful making sense of problems and persevering to solve them (MP. 1), reasoning abstractly and quantitatively (MP. 2), using appropriate tools strategically (MP.5), looking for and making use of structure (MP.7), and/or looking for and expressing regularity in repeated reasoning (MP.8).
- Score Point Distribution
 - 1.5% of students earned 6 points.
 - 4.9% of students earned 5 points.
 - 4.1% of students earned 4 points.
 - 4.6% of students earned 3 points.
 - 8.8% of students earned 2 points.
 - 16.7% of students earned 1 point.
 - 59.5% of students earned 0 points.

Score Rubric – Part A

Points	Attributes
3	Student response includes each of the following 3 elements. • Modeling component: Valid work or explanation to determine the amount of material needed for 1 pencil case • Computation component: Correct amount of material, in square inches, needed to make 1 pencil case, 80.68 • Modeling component: Valid expression that can be used to determine the amount of material, in square inches, to make n pencil cases, $80.68n$
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Sample Student Response:	Sample Solution 1: To find the sides, $7 \times 3 \times 2 = 42$, so both sides have a total surface area of 42. Next, to find the triangles (top and bottom), $\frac{3 \times 3}{2} \times 2$, so 9 in total. Finally, to find the middle, $4.24 \times 7 = 29.68$. So, in total $42 + 9 + 29.68 = 80.68$ sq. in. The amount of material needed to make one pencil case is 80.68 square inches. To find an equation that determines the amount of material in square inches to make n pencil cases you use the equation $80.68 \times n$.
Annotation for Sample Student Response:	Solution 1, Score Point 3 The response receives full credit. It includes each of the 3 required elements. Modeling Component: • Student Response: To find the sides, $7 \times 3 \times 2 = 42$, so both sides have a total surface area of 42. Next, to find the triangles (top and bottom), $\frac{3 \times 3}{2} \times 2$, so 9 in total. Finally, to find the middle, $4.24 \times 7 = 29.68$. So, in total $42 + 9 + 29.68 = 80.68$ ○ Rationale for Score: Valid work provided to determine the amount of material need to make 1 pencil case. The student accounts for each of the three rectangle areas and the two triangle areas; then adds the areas and finds a total amount of material needed (To find the sides, $7 \times 3 \times 2 = 42$. . . find the triangles (top and bottom), $\frac{3 \times 3}{2} \times 2$, so 9 in total . . . to find the middle, $4.24 \times 7 = 29.68$. . . $42 + 9 + 29.68 = 80.68$ sq. in.). Computation Component: • Student Response: The amount of material needed to make one pencil case is 80.68 square inches. ○ Rationale for score: A correct amount of material needed to make 1 pencil case is provided (80.68 square inches). Modeling Component: • Student Response: To find an equation that determines the amount of material in square inches to make n pencil cases you use the equation $80.68 \times n$. ○ Rationale for score: A valid expression is provided that can be used to determine the amount of material needed to make n pencil cases ($80.68 \times n$). Note: Sample student responses are not representative of all correct answers for an item and are only provided as a guide to assist teachers with scoring.

Scoring Rubric – Part B

Points	Attributes
3	Student response includes each of the following 3 elements. • Computation component: Correct area, in square inches, for one piece of material, 144 • Computation component: Correct cost, per square inch, of the material, \$0.03 • Modeling component: Valid work or explanation to find the cost per square inch of the material
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Sample Student Response:	Sample Solution 1: 1 foot is 12 inches so that means that these squares of material are 12 inches by 12 inches. Using the equation 12×12 you get the area of the square which is 144 square inches. To find the cost per square inch, you know that 144 square inches costs \$5.00, so the equation $5 \div 144 = 0.03472$ finds the cost per square inch which you would round to 0.03. It costs \$0.03 for 1 square inch of material.
Annotation for Sample Student Response:	Solution 1, Score Point 3 The response receives full credit. It includes each of the 3 required elements. Computation Component: • Student Response: 1 foot is 12 inches so that means that these squares of material are 12 inches by 12 inches. Using the equation 12×12 you get the area of the square which is 144 square inches. ○ Rationale for Score: A correct area, in square inches, for one piece of material is provided (144 square inches). Computation Component: • Student Response: It costs \$0.03 for 1 square inch of material. ○ Rationale for score: A correct cost per square inch of material is provided (\$0.03). Modeling Component: • Student Response: To find the cost per square inch, you know that 144 square inches costs \$5.00, so the equation $5 \div 144 = 0.03472$ finds the cost per square inch which you would round to 0.03. ○ Rationale for score: Valid work is provided to find the cost per square inch of the material. The student divides the cost of one piece of material by the total square inches in one piece, to find the cost per square inch (144 square inches costs \$5.00, so the equation $5 \div 144 = 0.03472$ finds the cost per square inch which you would round to 0.03). Note that rounding to the hundredth position shows understanding that this is a monetary value. Note: Sample student responses are not representative of all correct answers for an item and are only provided as a guide to assist teachers with scoring.

ANSWER KEY: ITEM SET 3

Item Set 3 – Question 1 (Equation Editor)

A turtle travels $7\frac{7}{8}$ miles in $\frac{3}{7}$ of a week. At this rate, how many miles per week will the turtle travel?

Enter your answer in the space provided. Enter **only** your answer.

$18\frac{3}{8}$

Item Information

- Answer – See Image, or an equivalent number.
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.RP.A.1
 - Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities measured in like or different units. For example, if a person walks $\frac{1}{2}$ mile in each $\frac{1}{4}$ hour, compute the unit rate as the complex fraction $\frac{1/2}{1/4}$ miles per hour, equivalently 2 miles per hour.
- Evidence Statement – 7.RP.1
 - Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. For example, if a person walks $\frac{1}{2}$ mile in each $\frac{1}{4}$ hour, compute the unit rate as the complex fraction $\frac{1/2}{1/4}$ miles per hour, equivalently 2 miles per hour. i) Tasks have a real-world context. ii) An example of like units would be cups of butter to cups of sugar; an example of different units would be miles to hours. Unit conversion is not assessed here.
- Subclaim – A – Major Content
 - The student solves problems involving the Major Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.129

Item Set 3 – Question 2 (Constructed Response)

A teacher creates a grading system for a 5-week period.

- There is a weekly quiz worth 18 points.
- There is a weekly homework assignment worth a fixed number of points.
- The total points for all homework assignments during the 5-week period is 170 points.

The equation $5(x + 18) = 170$ represents this situation.

- Show or explain each step to solve for x .
- State the value of x .
- Explain what the solution for x represents in this situation.

Enter your explanations and your answer in the space provided.

Item Information

- Answer – See Scoring Rubric and Sample Response
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.EE.B.4.a
 - Solve word problems leading to equations of the form $px (+/-) q = r$ and $p(x (+/-) q) = r$, where p , q and r are specific rational numbers. Solve equations of these forms fluently. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. For example, the perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?
- Evidence Statement – 7.C.5
 - Given an equation, present the solution steps as a logical argument that concludes with the set of solutions (if any). Content Scope: Knowledge and skills articulated in 7.EE.4a.
- Subclaim – C – Expressing Mathematical Reasoning
 - The student expresses grade/course-level appropriate mathematical reasoning by constructing viable arguments, critiquing the reasoning of others, and/or attending to precision when making mathematical statements.
- Score Point Distribution
 - 10.1% of students earned 3 points.
 - 15.6% of students earned 2 points.
 - 13.7% of students earned 1 point.
 - 60.6% of students earned 0 points.

Points	Attributes
3	Student response includes the following 3 elements. • Reasoning Component =1 point: Valid explanation or steps shown on how to solve the equation for x. • Computation Component =1 point: Correct solution to the equation, $x=16$. • Reasoning Component =1 point: Valid explanation for what the solution for x means for this situation. Sample Response: To solve for x, divide both sides of the equation, $5(x + 18) = 170$, by 5, so $x + 18 = 34$. Subtract 18 from both sides of the equation, so $x = 16$. The solution $x=16$ means each weekly homework assignment is worth 16 points for this 5-week grading period.
2	Student response includes 2 of the 3 elements from the 3-point description. Note: If the solution for x is incorrect, but the interpretation of the solution for x is correct, then 1 reasoning point is earned.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Sample Student Response:	Sample Solution 1: $5(x + 18) = 170$ $5x + 90 = 170$ subtract 90 from both sides $5x = 80$ $5x \div 5 = 80 \div 5$ $x = 16$ The solution for x (16) is how many points each homework assignment is worth over the 5 week grading period.
Annotation for Sample Student Response:	Solution 1, Score Point 3 The response receives full credit. It includes each of the 3 required elements. Reasoning Component: • Student Response: ○ Rationale for Score: Valid work is provided to solve the equation for x [$5(x + 18) = 170$, $5x + 90 = 170$, subtract 90 from both sides, $5x = 80$, $5x \div 5 = 80 \div 5$, $x = 16$]. The response correctly shows the equation written algebraically, then correctly uses the distribution property and solves for the variable x. The solution process must show proper use of a variable to solve the equation to receive credit for this element. Computation Component: • Student Response: $x = 16$ ○ Rationale for score: Correct value of x is given ($x = 16$). Reasoning Component: • Student Response: x (16) is how many points each homework assignment is worth ○ Rationale for score: Valid explanation for what the value of x represents is provided (The solution for x (16) is how many points each homework assignment is worth over the 5 week grading period). Note: Sample student responses are not representative of all correct answers for an item and are only provided as a guide to assist teachers with scoring.

Item Set 3 – Question 3 (Selected Response)

A teacher has an equal number of pink, blue, green, yellow, and purple erasers in a box. She will randomly distribute 1 eraser from the box to each student.

What is the probability that the first eraser that will be distributed is a color other than yellow?

- ☐ A. $\frac{1}{5}$
- ☐ B. $\frac{1}{4}$
- ☐ C. $\frac{3}{4}$
- ☒ D. $\frac{4}{5}$

Item Information

- Answer – D
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.SP.C.7.a
 - Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. For example, if a student is selected at random from a class, find the probability that Jane will be selected and the probability that a girl will be selected.
- Evidence Statement – 7.SP.7a
 - Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy. a. Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. For example, if a student is selected at random from a class, find the probability that Jane will be selected and the probability that a girl will be selected. i) Simple events only.
- Subclaim – B – Supporting Content
 - The student solves problems involving the Additional and Supporting Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.407

Item Set 3 – Question 4 (Constructed Response)

A person is planning an event and wants to order chicken and beans from a store.

- The store charges \$9.49 per pound of chicken and \$5.75 for one quart of beans.
- The person has \$70 to purchase the chicken and the beans.

Let p represent the number of pounds of chicken.

- Create an equation that can determine the maximum amount of chicken, in pounds, that can be purchased, along with one quart of beans, using the \$70. Do not include tax.
- Solve the equation to determine the maximum amount of chicken that can be purchased. Round your final answer to the nearest quarter pound. Show your work.

Enter your equation, your solution, and your work in the space provided.

Item Information

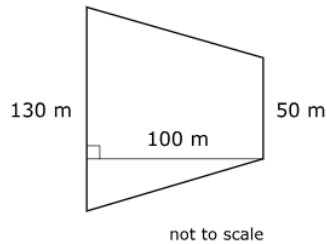
- Answer – See Scoring Rubric and Sample Student Responses
- Colorado Academic Standards (CAS) Evidence Outcomes
 - 7.EE.B.4.a – Solve word problems leading to equations of the form $px (+/-) q = r$ and $p(x (+/-) q) = r$, where p , q and r are specific rational numbers. Solve equations of these forms fluently. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. For example, the perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?
 - 7.EE.B.3 – Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. For example: If a woman making \$25 an hour gets a 10% raise, she will make an additional $\frac{1}{10}$ of her salary an hour, or \$2.50, for a new salary of \$27.50. If you want to place a towel bar $9\frac{3}{4}$ inches long in the center of a door that is $27\frac{1}{2}$ inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.
- Evidence Statement – 7.D.1
 - Solve multi-step contextual word problems with degree of difficulty appropriate to Grade 7, requiring application of knowledge and skills articulated in Type I, Sub-Claim A Evidence Statements. i) Tasks may have scaffolding, if necessary, in order to yield a degree of difficulty appropriate to Grade 7. ii) Tasks involving writing or solving an equation should not go beyond the equation types described in 7.EE.4a. [$px + q = r$ and $p(x + q) = r$ where p , q , and r are specific rational numbers.]
- Subclaim – D – Modeling and Application
 - The student solves real-world problems with a degree of difficulty appropriate to the grade/course by applying knowledge and skills articulated in the standards for the current grade/course (or for more complex problems, knowledge and skills articulated in the standards for previous grades/courses), engaging particularly in the Modeling practice, and where helpful making sense of problems and persevering to solve them (MP. 1), reasoning abstractly and quantitatively (MP. 2), using appropriate tools strategically (MP.5), looking for and making use of structure (MP.7), and/or looking for and expressing regularity in repeated reasoning (MP.8).
- Score Point Distribution
 - 5.0% of students earned 3 points.
 - 11.7% of students earned 2 points.
 - 15.0% of students earned 1 point.
 - 68.3% of students earned 0 points.

Points	Attributes
3	Student response includes each of the following 3 elements. • Modeling component: Valid equation equivalent to $9.49p + 5.75 = 70$. • Computation component: Correct pounds of chicken that can be purchased, rounded to the nearest quarter pound, 6.75. • Modeling component: Valid work to solve the given equation and determine the maximum amount of chicken, in pounds, that can be purchased.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Sample Student Response:	Equation: $9.49p + 5.75 = 70$ Maximum amount of chicken: 6 and $\frac{3}{4}$ pounds Your Work: $70 - 5.75 = 64.25$ $64.25 \div 9.49 = p$ $p = 6$ and $\frac{3}{4}$
Annotation for Sample Student Response:	Score Point 3 The response receives full credit. It includes each of the 3 required elements. Modeling Component: • Student Response: Equation: $9.49p + 5.75 = 70$. ○ Rationale for Score: The student provides a valid equation that determines the maximum amount of chicken, in pounds, that can be purchased, along with one quart of beans, using 70 dollars ($9.49p + 5.75 = 70$). The variable, p, is used to represent the number of pounds that can be purchased. Note that variable substitution is allowed. Computation Component: • Student Response: Maximum amount of chicken: 6 and $\frac{3}{4}$ pounds ○ Rationale for score: The correct pounds of chicken, rounded to the nearest quarter pound, is provided (6 and $\frac{3}{4}$). Modeling Component: • Student Response: Your Work: $70 - 5.75 = 64.25$, $64.25 \div 9.49 = p$, $p = 6$ and $\frac{3}{4}$ ○ Rationale for score: The student provides valid work for how to solve the equation to determine the maximum pounds of chicken that can be purchased ($70 - 5.75 = 64.25$, $64.25 \div 9.49 = p$, $p = 6$ and $\frac{3}{4}$). Note: The equation must have one variable and at least one operation. Note: Equation should not include tax. Note: Variable substitution is allowed. Note: Sample student responses are not representative of all correct answers for an item and are only provided as a guide to assist teachers with scoring.

Item Set 3 – Question 5 (Fill in the Blank)

Part A

A piece of land is in the shape of a trapezoid, with parallel sides that are 50 meters and 130 meters long. The distance between the parallel sides is 100 meters, as shown in the diagram.

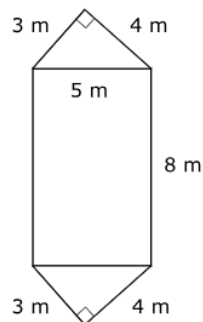


How large, in square meters, is the piece of land?

Enter your answer in the box.

Part B

There is a large picnic area on the property. It is made up of a rectangle, with dimensions of 5 meters by 8 meters, and two right triangles, whose legs are 3 meters and 4 meters long, as shown in the diagram.



How large, in square meters, is the picnic area?

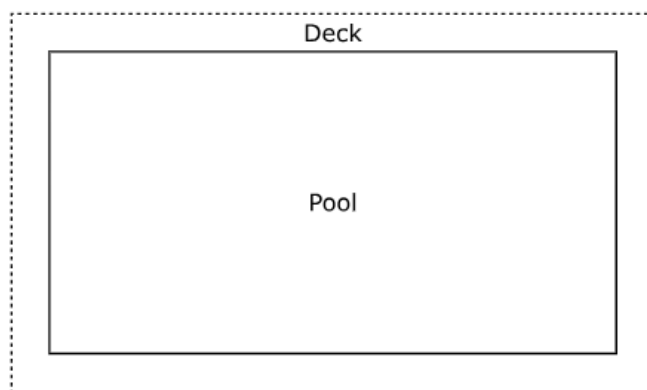
Enter your answer in the box.

Item Information

- Part A Answer – See Image
- Part B Answer – See Image
- Colorado Academic Standards (CAS) Evidence Outcomes – 7.G.B.6
 - Solve real-world and mathematical problems involving area, volume, and surface area of two- and three-dimensional objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms.
- Evidence Statement – 7.G.6
 - Solve real-world and mathematical problems involving area, volume, and surface area of two- and three-dimensional objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms. i) Pool should contain tasks with and without context. ii) Tasks focus on area of two-dimensional objects.
- Subclaim – B – Supporting Content
 - The student solves problems involving the Additional and Supporting Content for her grade/course with connections to the Standards for Mathematical Practice.
- P Value – 0.149

Item Set 3 – Question 6 (Selected Response, Constructed Response)

A person is building a deck around a rectangular pool, as shown in the diagram.



- The width, W , of the pool is 8 feet, and the length, L , of the pool is 15 feet.
- The expression $2(W + 12) + 2(L + 12)$ can be used to find the total perimeter of the deck, in feet.
- The person uses a different expression, $2W + 2L + 48$, to find the perimeter of the deck.

Part A

How many terms are in the expression $2W + 2L + 48$?

- ☐ A. 2
- ☒ B. 3
- ☐ C. 4
- ☐ D. 5

Part B

- Explain whether the two expressions for finding the perimeter of the deck are equivalent.
- The person claims the perimeter of the deck will be twice the perimeter of the pool. Explain whether the person's claim is correct or incorrect.
- State the perimeter of the pool and the perimeter of the deck.

Enter your explanations and your answers in the space provided.

Item Information

- Part A Answer – B
- Part B Answer – See Scoring Rubric and Sample Student Responses
- Colorado Academic Standards (CAS) Evidence Outcomes
 - 6.EE.A.4 – Identify when two expressions are equivalent (i.e., when the two expressions name the same number regardless of which value is substituted into them). For example, the expressions $y + y + y$ and $3y$ are equivalent because they name the same number regardless of which number y stands for.
 - 6.EE.B.6 – Use variables to represent numbers and write expressions when solving a real-world or mathematical problem; recognize that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set.
 - 6.EE.A.2.b – Identify parts of an expression using mathematical terms (sum, term, product, factor, quotient, coefficient); view one or more parts of an expression as a single entity. For example, describe the expression $2(8 + 7)$ as a product of two factors; view $(8 + 7)$ as both a single entity and a sum of two terms.
- Evidence Statement – 7.C.8
 - Construct, autonomously, chains of reasoning that will justify or refute propositions or conjectures. Content Scope: Knowledge and skills articulated in 6.NS.C, 6.EE.A, 6.EE.B. i) Tasks may have scaffolding, if necessary, in order to yield a degree of difficulty appropriate to Grade 7.
- Subclaim – C – Expressing Mathematical Reasoning
 - The student expresses grade/course-level appropriate mathematical reasoning by constructing viable arguments, critiquing the reasoning of others, and/or attending to precision when making mathematical statements.
- Score Point Distribution
 - 2.6% of students earned 4 points.
 - 5.9% of students earned 3 points.
 - 13.5% of students earned 2 points.
 - 35.1% of students earned 1 point.
 - 42.9% of students earned 0 points.

Scoring Rubric – Part A

Points	Attributes
1	Student response is $\frac{5}{6}x - 1$ or an equivalent expression with exactly two terms. Variable substitution is not allowed. $\frac{1}{2}x + 3 + \frac{1}{3}x - 4$ $\frac{3}{6}x + 3 + \frac{2}{6}x - 4$ $\frac{5}{6}x - 1$
0	The response is incorrect or irrelevant.

Scoring Rubric – Part B

Points	Attributes
3	Student response includes the following 3 elements. Reasoning component = 1 point: Valid explanation on how the expression $2(W + 12) + 2(L + 12)$ is equivalent to $2W + 2L + 48$. Reasoning component = 1 point: Valid explanation on why the person is incorrect in claiming the perimeter of the deck will be twice the perimeter of the pool. Computation component = 1 point: The student includes the correct perimeter of the pool, 46 feet, and the deck, 94 feet. Sample Student Response: Both expressions are equivalent. If I use the distributive property, I can rewrite the expression $2(W + 12) + 2(L + 12)$ as $2W + 24 + 2L + 24$, which then I can combine the like terms so that the expression is $2W + 2L + 48$. The person is incorrect. The perimeter of the pool is $15 + 15 + 8 + 8 = 46$ feet. Twice the perimeter of the pool is $46 \times 2 = 92$. The perimeter of the deck is $2(15) + 2(8) + 48 = 94$ feet. Therefore, the person's claim is incorrect.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Sample Student Response:	Sample Solution 1: The two expressions are equivalent. $2W + 2L + 48$ is the simplified version of $2(W + 12) + 2(L + 12)$. You just distribute the numbers. $2(W + 12) + 2(L + 12)$ $2W + 24 + 2L + 24$ $2W + 2L + 48$ The person's claim is incorrect. The width, w, is 8 and the length, l, is 15. To compare the two perimeters, you have to calculate them first. Deck: $2(8) + 2(15) + 48$ $16 + 30 + 48$ 94 Pool: $P = 2L + 2W$ $P = 2(15) + 2(8)$ $P = 30 + 16$ $P = 46$ So the perimeter of the deck is 94 feet, and the perimeter of the pool is 46 feet. 46 times 2 is 92, not 94, so the deck's perimeter is not twice more than the perimeter of the pool.
Annotation for Sample Student Response:	Solution 1, Score Point 3 The response receives full credit. It includes each of the 3 required elements. Reasoning Component: Student Response: You just distribute the numbers. $2(W + 12) = 2(L + 12)$, $2W + 24 + 2L + 24$, $2W + 2L + 48$ Rationale for Score: A valid explanation for how the two expressions are equivalent is provided [The two expressions are equivalent. $2W + 2L + 48$ is the simplified version of $2(W + 12) + 2(L + 12)$. You just distribute the numbers. $2(W + 12) = 2(L + 12)$, $2W + 24 + 2L + 24$, $2W + 2L + 48$]. Reasoning Component: Student Response: perimeter of the deck is 94 feet, and the perimeter of the pool is 46 feet. 46 times 2 is 92, not 94, so the decks perimeter is not twice more than the perimeter of the pool Rationale for score: A valid explanation for why the person is incorrect in their claim

	that the perimeter of the deck will be twice the perimeter of the pool is provided (The persons claim is incorrect . . . the perimeter of the deck is 94 feet, and the perimeter of the pool is 46 feet. 46 times 2 is 92, not 94, so the decks perimeter is not twice more than the perimeter of the pool). The student provides correct perimeters for the pool and the deck, which are then correctly used to compare the two perimeters and support that the person’s claim is incorrect. Computation Component: • Student Response: deck is 94 feet, pool is 46 feet ○ Rationale for score: A correct perimeter of the deck and the pool are provided (deck is 94 feet, pool is 46 feet). Note that label of ‘feet’ is provided in the prompt and is not required to be added to the students answer. No work for how the perimeters were calculated is required in order to receive credit for this element. Note: Sample student responses are not representative of all correct answers for an item and are only provided as a guide to assist teachers with scoring.
--	--